From

2018/01/01

To

2026/04/06

📊 REGIME MODEL

DMA Window (days)

200

HMM Iterations

500

Forecast Horizon (days)

60

Macro features in HMM

📈 STRATEGY

MA Window (days)

5

Transaction Cost (bps)

5

🖥 PANELS

- Regime Shading
- DMA Line
- Macro Panel
- Yield Curve Panel
- Volatility Panel
- Drawdown Panel
- Live Mode (60s)

Refresh Data

Fetches latest Nifty 50 from yfinance.

v5 Regime Multipliers

- Bull Quiet → 1.00×
- Bull Volatile → 1.00×
- Neutral → 0.75×
- Stress → 0.35×
- Panic → 0.00×

NAUTILUS V0.5.0

MACRO-REGIME INTELLIGENCE · 5-STATE HMM · NO LOOK-AHEAD · RESEARCH USE ONLY

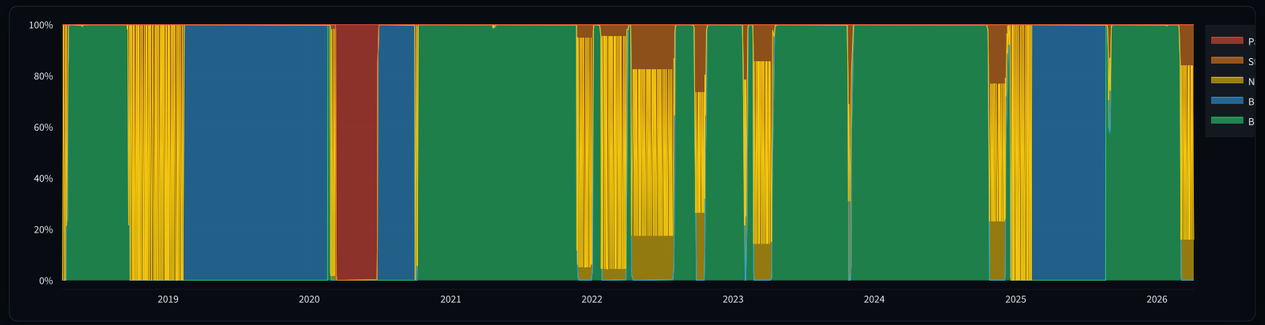
NIFTY 50 · ^GSPC · ^TNX · ^IRX

Nifty 50	200D MA	HMM Regime	Soft Kelly	10Y G-Sec	RBI Repo Rate	US 10Y–3M Spread
22,668	25,196	Stress (84…	0.41×	7.129%	6.00%	+0.71pp
↓ -0.20% 1D	↓ -10.0% spread			↑ +0bps 1D		

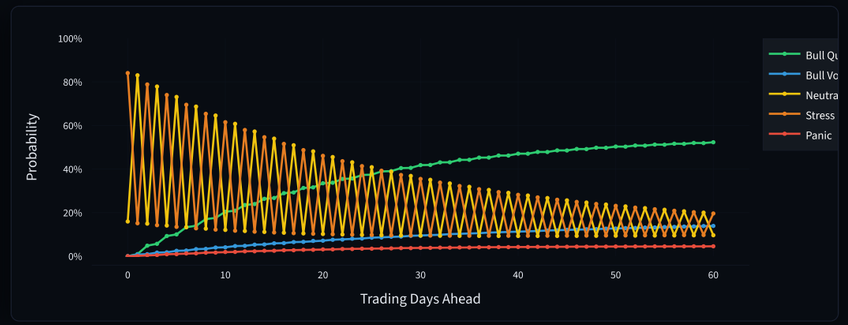
Bull Quiet 52% · Bull Volatile 23% · Neutral 11% · Stress 11% · Panic 4%

NIFTY 50 · REGIME SHADING · SOFT KELLY

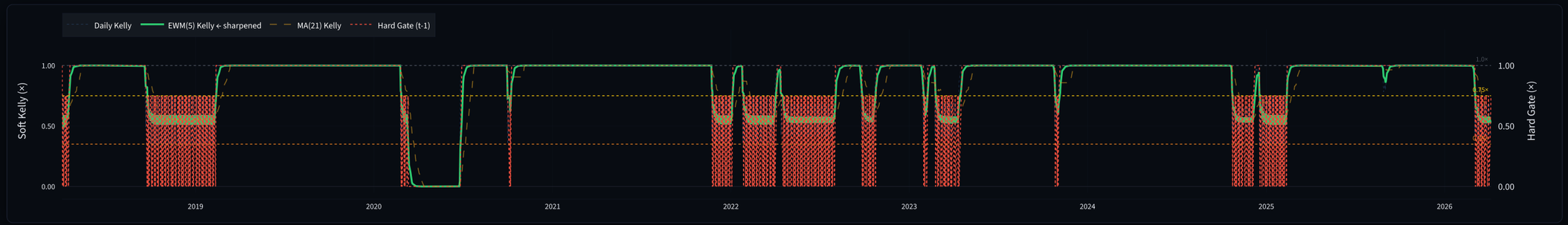
REGIME PROBABILITY STACK



REGIME FORWARD FORECAST



SOFT KELLY — SHARPENED SIGNAL HISTORY



EWM(5) = exponentially-weighted Kelly with 5-day half-life — ~2× faster response vs MA(21). Hard Gate (right axis, dotted red) = binary 1.0/0.75/0 from dominant HMM state, pre-shifted +1D.

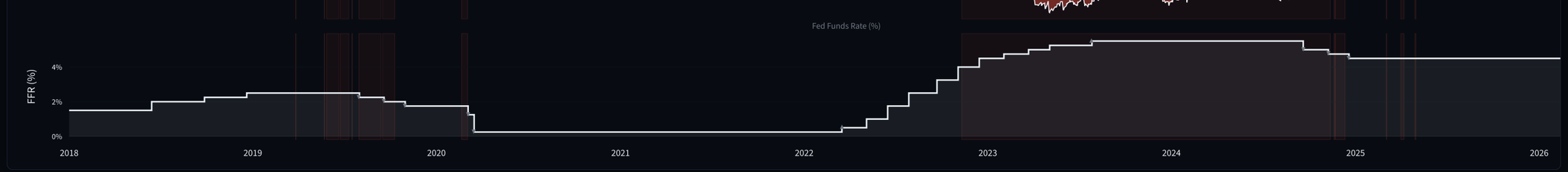
MONTHLY RETURNS HEATMAP

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual
2026	-3.1%	-0.6%	-11.3%	1.5%									-13.2%
	-0.6%	-5.9%	6.3%	3.5%	1.7%	3.1%	-2.9%	-1.4%	0.8%	4.5%	1.9%	-0.3%	10.5%
2024	-0.0%	1.2%	1.6%	1.2%	-0.3%	6.6%	3.9%	1.1%	2.3%	-6.2%	-0.3%	-2.0%	8.8%
	-2.4%	-2.0%	0.3%	4.1%	2.6%	3.5%	2.9%	-2.5%	2.0%	-2.8%	5.5%	7.9%	20.0%
2022	-0.1%	-3.1%	4.0%	-2.1%	-3.0%	-4.8%	8.7%	3.5%	-3.7%	5.4%	4.1%	-3.5%	4.3%
	-2.5%	6.6%	1.1%	-0.4%	6.5%	0.9%	0.3%	8.7%	2.8%	0.3%	-3.9%	2.2%	24.1%
2020	-1.7%	-6.4%	-23.2%	14.7%	-2.8%	7.5%	7.5%	2.8%	-1.2%	3.5%	11.4%	7.8%	14.9%
	-0.3%	-0.4%	7.2%	1.5%	1.5%	-1.1%	-5.7%	-0.9%	4.1%	3.5%	1.5%	0.9%	12.0%
2018		-4.9%	-3.6%	6.2%	-0.0%	-0.2%	6.0%	2.9%	-6.4%	-5.0%	4.7%	-0.1%	-1.5%

MACRO PANEL — RBI REPO RATE & 10Y G-SEC YIELD

RBI Policy Events: 2020-03-27 COVID cut −75bps · 2020-05-22 COVID cut −40bps · 2022-05-04 Hike +40bps · 2022-06-08 Hike +50bps · 2022-08-05 Hike +50bps · 2023-02-08 Final hike +25bps · 2025-02-07 Cut −25bps · 2025-04-09 Cut −25bps

US MACRO MONITOR — S&P 500 · 10Y−3M TREASURY SPREAD · FED FUNDS RATE



Blue shading in SPX panel = yield curve inversion period (10Y < 3M). Red fill = inverted spread — leading recession signal. ▲▼ on FFR panel = Fed policy moves. Data: ^GSPC, ^TNX, ^IRX via yfinance; FFR from policy anchors.

REALISED VOLATILITY REGIME

21D Realised Vol	63D Realised Vol	Vol Ratio (21/63)	Vol Regime
25.5%	17.8%	1.43×	High Vol
↑ +7.7pp vs 63D			

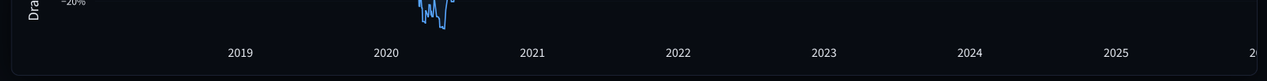
DRAWDOWN FROM PEAK

Current Drawdown	Max Drawdown (period)
-13.9%	-38.4%

BACKTEST — MA(5) VS REGIME-FILTERED (V5 HARD GATE)

Metric	MA(5) Only	+ Regime Gate (v5)	Buy & Hold
Total Return	45.5%	54.5%	117.1%
CAGR	4.6%	5.4%	9.8%
Sharpe	0.47	0.69	0.65
Sortino	0.43	0.71	0.78
Max DD	-26.5%	-12.0%	-38.4%
Calmar	0.18	0.45	0.26
Win Rate	32.1%	28.2%	54.1%

Hard gate (v5): Bull 1.0×, Neutral 0.75×, Stress 0×, Panic 0×.
Tx cost 5bps/side. All signals pre-shifted +1D (no look-ahead).



REGIME ATTRIBUTION & STATISTICS

Regime	Days	% Time	Ann. Return	Sharpe	Avg Day	Max DD	Gate
Bull Quiet	1032	52.3%	18.0%	1.37	0.069%	-11.8%	1.00×
Bull Volatile	445	22.5%	15.4%	1.07	0.061%	-11.4%	1.00×
Neutral	212	10.7%	-29.7%	-1.84	-0.133%	-25.8%	0.75×
Stress	214	10.8%	23.3%	1.31	0.089%	-10.5%	0.35×
Panic	71	3.6%	-5.4%	0.16	0.033%	-27.2%	0.00×

Regime Transition Matrix

LIVE SIGNAL PREVIEW

Raw Signal

0.70

Soft Kelly

0.414×

Hard Gate

0.75×

Gated Signal

0.500

↑ Δ-0.200

Hard gate applied: `0.75` → signal `0.70` → `0.500`

DATA EXPORTS

⬇ Signal Table (CSV) | ⬇ Backtest Summary (CSV) | ⬇ Equity Curves (CSV)

Methodology — Model Design, Data Sources & No Look-Ahead

v5 Architecture — Hard Regime Gates

```
position(t) = base_signal(t) × hard_gate(t-1)

hard_gate(t) = argmax P(state | data_{1..t}) → mult
  🟢 Bull Quiet    = 1.00   🔵 Bull Volatile = 1.00
  🟡 Neutral       = 0.75
  🟠 Stress        = 0.00   🔴 Panic         = 0.00
```

The v5 hard gate produces a **measurable difference** vs the base MA signal by zeroing positions entirely in Stress/Panic regimes. Soft Kelly (continuous weighted multiplier) is shown for reference but the backtest uses the hard gate.

HMM Features (`vol_21d, ret_21d, vol_ratio, vol_of_vol, ret_5d, drawdown, bond_yield_chg_21d, yield_spread, repo_easing, dma_200_ratio`)

Group	Variables
Price/Vol	21D vol, 21D ret, vol ratio 21/63, vol-of-vol, 5D ret, 252D drawdown
Macro	10Y yield Δ21D, yield spread, RBI easing flag, 200-DMA ratio

Data Sources

Series	Source	Coverage
Nifty 50	yfinance `^NSEI`	Live
RBI Repo Rate	`data/rbi_repo_rate.csv` (RBI)	2012→present
India 10Y G-Sec	`data/india_10y_yield.csv` (Investing.com)	2018→present
US 10Y Treasury	yfinance `^TNX`	Live
US 3M T-Bill	yfinance `^IRX`	Live

No Look-Ahead

All signals and gates shifted +1D. Backtest engine takes pre-shifted signals.